a Biological Model and Silicone Biofilm Substrate

Mycobacterium marinum 1218R

RFP

BSL-2 surrogate for Mycobacterium tuberculosis

Stable chromosomal RFP at Giles phage attB site

Culture Vessel Specifications (31°C, 4 Days):

- Polydimethylsiloxane (PDMS) silicone membranes
- Suspended biofilm-forming cell pellets harvested for RNA-seq
- $n = 3$ biological replicates per experimental modality

b Simulated Microgravity and Ground Hardware

1. Lab-Designed 3D Clinostat ($n = 3$)

Continuous 2-axis clinorotation ($I = 1.5$ rpm, $O = 3.825$ rpm)

Samples: RFP3D11, RFP3D39, RFP3D47

2. Random Positioning Machine 2.0 ($n = 3$)

Random multi-axis velocity vectoring, time-averaged $< 0.01g$

Samples: RFPRPM4, RFPRPM41, RFPRPM6

3. Static 1g Earth Control ($n = 3$)

Stationary incubator shelf adjacent to simulators

Samples: RFPNG14, RFPNG35, RFPNG45

c Empirical Multi-Scale Analytical Framework

1. Empirical Quant

NASA OSDR S3 Raw Reads

5,510 Annotated Genes

kallisto Pseudoalignment

351–738 Real DEGs

2. WGCNA Modules

Soft-Threshold $\beta = 6$

Topological Overlap (TOM)

5 GOSlim Co-Expression Modules

k_{within} **Hub Centrality**

3. TabPFN AI (Nature 2025)

Prior-Data Transformer

Bayesian In-Context Priors

15 Topological Hub Features

88.9% Binary LOOCV

4. Systems Validation

GOSlim Over-Representation

Cellular Metabolic Model

Kinematic Radar Trajectories

RO-Crate & Zenodo Deposit